

A decorative graphic on the left side of the slide consisting of white lines and circles on a blue gradient background, resembling a circuit board or neural network structure.

MINI RESEARCH PROJECT

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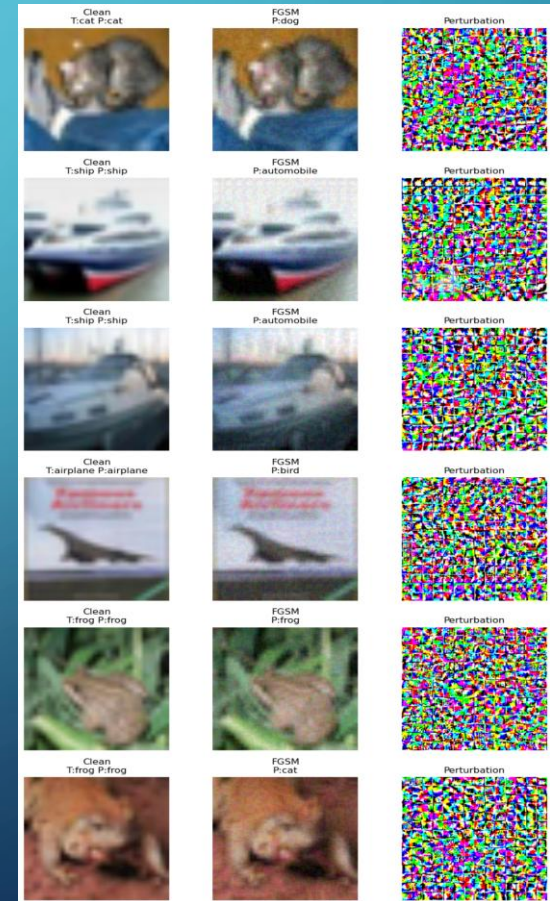
COURSE: ARTIFICIAL INTELLIGENCE

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INTRODUCTION

TITLE: ADVERSARIAL ROBUSTNESS AND INTERPRETABILITY IN VISION MODELS

- Deep learning models achieve high accuracy in image classification
- However, they are vulnerable to **adversarial perturbations**
- Small and imperceptible noise can cause **incorrect predictions**
- This poses risks in real-world applications (e.g., autonomous systems)



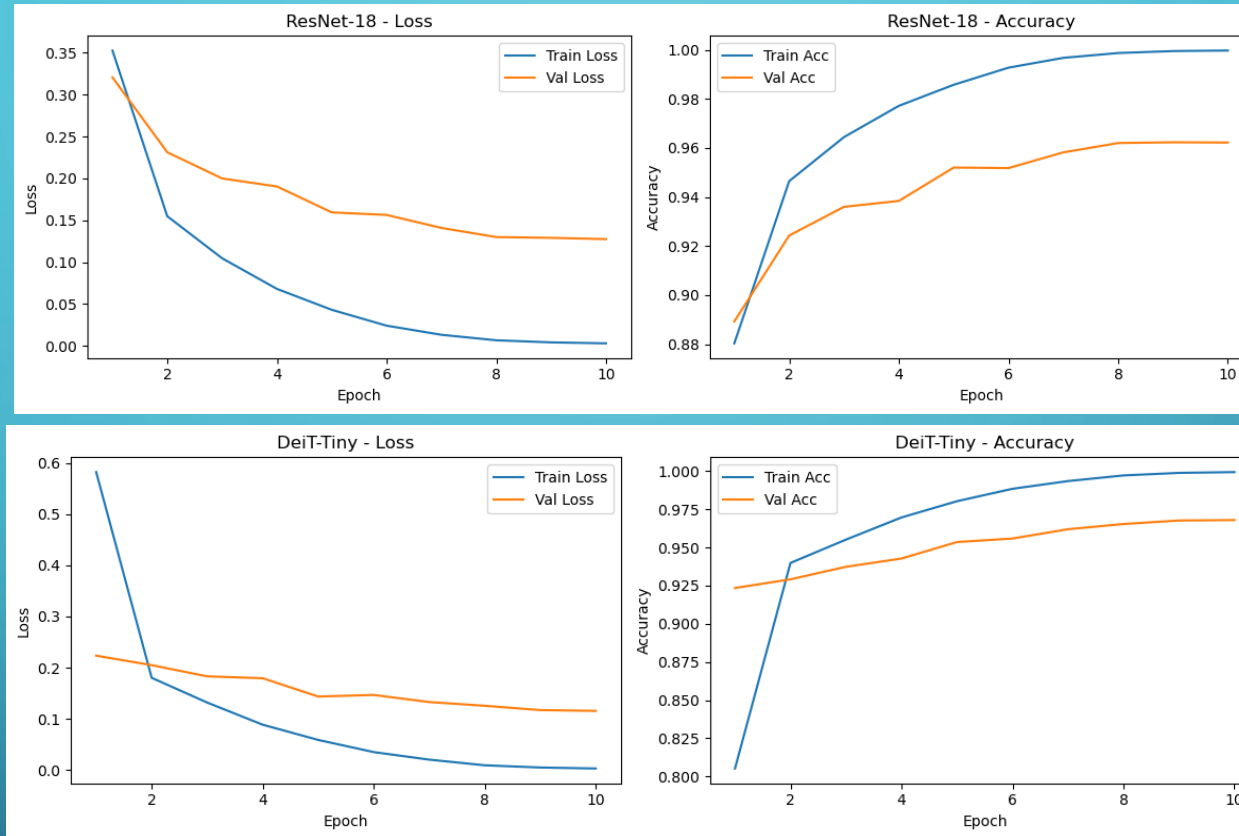
OBJECTIVES

- Compare two architectures:
 - **ResNet-18 (CNN)**
 - **DeiT-Tiny (Vision Transformer)**
- Evaluate robustness using:
 - FGSM (Fast Gradient Sign Method)
 - PGD (Projected Gradient Descent)
- Analyze performance under different perturbation strengths
- Use interpretability techniques:
 - Grad-CAM
 - Integrated Gradients

METHODOLOGY

- Dataset: **CIFAR-10 (10 classes)**
- Images resized to **224×224**
- Models trained using:
 - Cross-entropy loss
 - Adam optimizer
- Workflow:
 - Train models
 - Apply adversarial attacks
 - Evaluate performance
 - Visualize model behavior

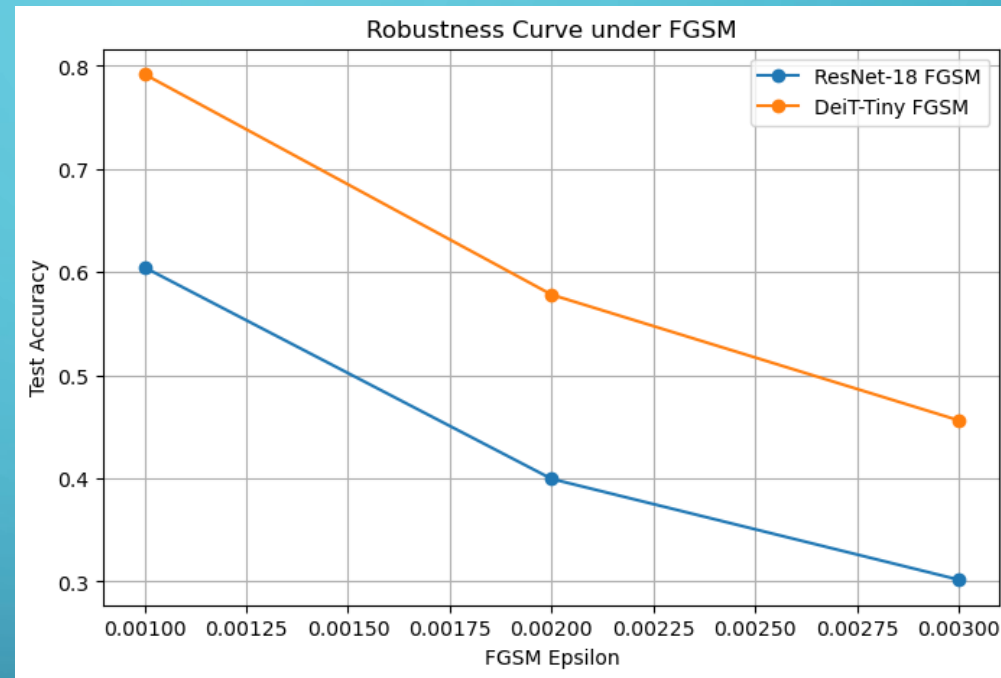
TRAINING AND PERFORMANCE



- Both models converge smoothly
- High validation accuracy
- DeiT slightly outperforms ResNet
- Minor overfitting observed (train > validation)

ROBUSTNESS ANALYSIS

- Accuracy decreases as perturbation (ϵ) increases
- FGSM causes significant degradation
- PGD (multi-step) is more destructive
- DeiT consistently outperforms ResNet under attack



Key Insight: Vision Transformers show higher robustness than CNNs under adversarial attacks

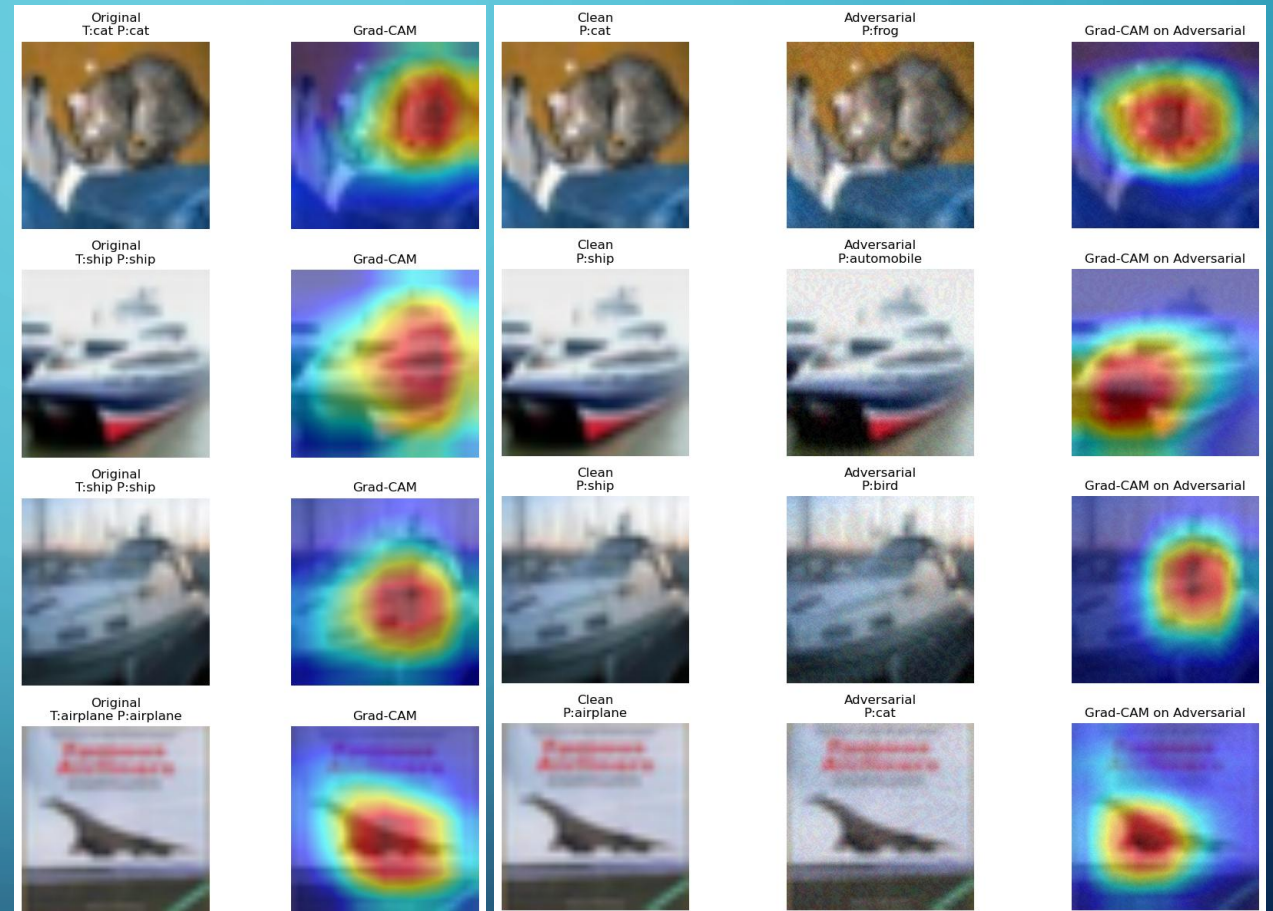
INTERPRETABILITY AND CONCLUSION

Interpretability:

- Grad-CAM highlights regions influencing predictions
- Adversarial inputs shift attention patterns

Conclusion:

- Small perturbations significantly impact performance
- CNNs are more vulnerable than Transformers
- Transformers capture more robust features
- Interpretability helps understand model decisions



“Even imperceptible noise can fool deep learning models”

REFERENCES

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